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Evaluating Natural Language Processing Algorithms for Improved Hazard and Operability Analysis

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ABSTRACT

Automating Hazard and Operability (HAZOP) reports is crucial for enhancing efficiency and reducing human biases in hazard identification process. Recent HAZOP studies have applied various Natural Language Processing (NLP) algorithms to optimize hazard identification and reporting. These studies exhibit divergent textual inputs and results and overlook the mining of all HAZOP report components. Furthermore, traditional NLP algorithms such as Bag of Words (BOW) and Term Frequency-inverse Document Frequency (TF-IDF) often fail to capture polysemy and semantic relationships between words. This study aims to evaluate NLP algorithms' efficacy in automating HAZOP reports to improve safety levels in infrastructure projects. These algorithms, including BOW, TF-IDF, the global vectors for word representation (GloVe), and sentence bidirectional encoder representation from transformers (SBERT), were combined with machine learning classifiers such as random forest (RF), gaussian naive bayes (NB), decision tree (DT), and k-nearest neighbors (KNN), using two small HAZOP datasets with varied inputs. A zero-shot text classification model was further evaluated for its ability to assign labels to HAZOP data without prior training. Results demonstrate that GloVe combined with RF achieves the highest accuracy (83 %), significantly outperforming other models. We further observe that KNN degrades on short text features, while DT underperforms on longer descriptions. The zero-shot model achieves low performance (52 % accuracy), lacking the precision needed for fine-grained, jargon-heavy labels. These findings indicate that GloVe embeddings remain a robust foundation for HAZOP automation under data-scarce conditions in the operations of infrastructure projects.

1. Introduction

Hazard and operability (HAZOP) study is the most widely used process hazard analysis (PHA) technique for hazard identification and analysis in the process industry (Zhou et al., 2018). In HAZOP analysis, participants begin by dividing the process flow diagram (PFD) or piping and instrument diagram (P&ID) into defined subsections, referred to as nodes, to ensure that each piece of equipment in the process is analyzed thoroughly. Through guided brainstorming sessions, participants employ guidewords (e.g., no, more, less of) in conjunction with process parameters (e.g., pressure, temperature, flow) to identify deviations (such as less flow, more temperature) from design intent or normal operations. This systematic approach, as illustrated in Figure 1, is applied within a specific node. Once deviations are identified, the team investigates their potential causes, consequences, safeguards, and recommendations to produce final HAZOP report (Dunjó et al., 2010). Since the HAZOP study was formulated, extensive endeavors have been conducted to enhance its efficacy in hazard identification. These initiatives involve deploying HAZOP across diverse industrial sectors, improvement of its procedures, integration with PHA tools, and different levels of automation (Elhosary and Moselhi, 2024). Research in HAZOP studies has primarily focused on the chemical and petrochemical engineering fields. However, it has been extended to new industries including environmental engineering for operator safety (Choi and Byeon, 2020) and various subareas of infrastructure projects to increase safety levels. These subareas encompass railway systems (Li et al., 2015), (Noorudheen et al., 2013), supply chain management (Mitkowski and Zenka-Podlaszewska, 2014), urban drainage system asset management (Stanić et al., 2014), demolition of large bulk materials (Joubert et al., 2021) and hoisting operations of concrete prefabricated buildings (Zhu et al., 2022). For example, the HAZOP method was applied to identify workplace accident risk factors and their consequences, highlighting the critical role of worker attitudes (Prima and Anthara, 2023). High-risk activities, comprising 38% of identified risks, included welding, electrical panels, and foreign objects in work areas, while moderate risks (7%) involved equipment like grinders and drilling machines. The study emphasized standardiz-

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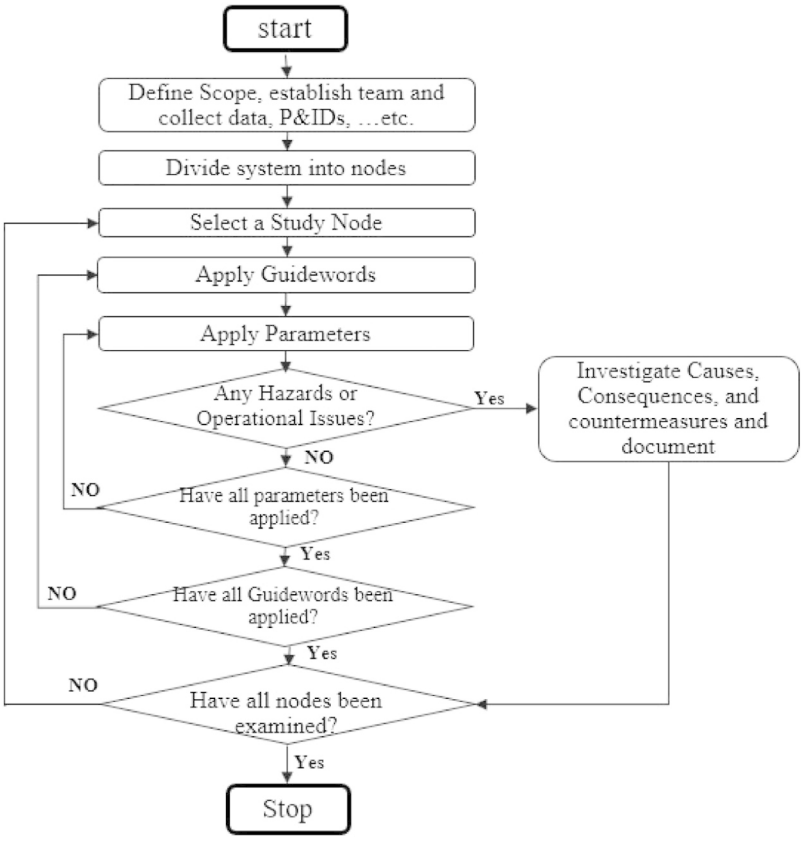


Figure 1. Flowchart of HAZOP study Procedures.

ing occupational safety procedures and promoting personal protective equipment (PPE) use to minimize accidents. A HAZOP study was conducted to assess technical and human factors in a safety system, leveraging an expert team to enhance track worker protection and reduce rail sector risks (Noorudheen et al., 2013). A safety risk assessment was performed for dismantling large bulk material handling machines, relying on project work method statements, construction regulations, and standards like AS2601: The Demolition of Structures, to formulate guide words (Joubert et al., 2021).

These studies underscore the efficacy of the HAZOP methodology in safeguarding workers and other project components against potential hazards and operational issues. However, they rely on traditional HAZOP procedures to generate reports to identify hazards and enhance safety levels, leading to notable challenges such as time-consuming process, resource-intensive efforts, and reliance on expert judgement. To tackle these issues and improve efficiency, there has been an increasing need to automate HAZOP reports within the operations of infrastructure projects. Indeed, automation of HAZOP study has been a research focus for over 30 years, with various knowledge-based approaches developed to support hazard identification, analysis, and mitigation in the process industry. These include expert systems (Kościelny et al., 2017), (Wu et al., 2023), (Hu et al., 2015), simulation-based tools (Yi et al., 2023), (Emami et al., 2022), integrated-based tools (Zenier and Antonello, 2023), (Bouafia et al., 2020), ontologies (Bai et al., 2021), (Yan et al., 2023), and Bayesian networks (Hu et al., 2012), (Wang et al., 2023). However, acquiring, eliciting, and updating domain-specific knowledge remains challenging, subjective, and time-consuming, relying heavily on expert input and iterative refinement (Elhosary and Moselhi, 2024). In the recent few years, data driven models, including machine learning technology, deep learning, and natural language processing (NLP) algorithms, have been used to automate various aspects of HAZOP reports by leveraging insights from historical data.

Automating HAZOP reports plays a significant role in reducing time, cost, and human bias in hazard identification processes. These reports encapsulate a wealth of knowledge and experience from the experts who contributed to their creation. For instance, information from HAZOP reports, such as the causes, safeguards, and recommendations, can serve as a valuable reference for accident prevention. Consequently, there is a growing demand to extract useful information from HAZOP reports. The expertise and knowledge of safety experts, presented in textual form within HAZOP reports, can be learned and transferred to similar fields using natural language processing (NLP). Notably, despite the advancements in automation across various industries, automation remains largely unexplored in HAZOP study within infrastructure projects. The repetitive nature of operations in infrastructure projects underscores the potential for automating aspects (like predicting deviations, consequences, safeguards, and recommendations) of HAZOP study. This can reduce both subjectivity and the time required for HAZOP sessions, particularly when participants have limited experience due to staff turnover and experts' retirement. Therefore, this area, particularly NLP, has captured our attention to explore how its algorithms can be leveraged to automate HAZOP reports in the operations of infrastructure projects, even in cases of limited datasets. In a relevant study, Bag of Words (BOW) and machine learning (ML) classifiers were employed to convert causes (42 samples) into feature vectors, achieving 92% accuracy in predicting deviations within HAZOP study worksheets for a Tankage Area case study in the chemical process industry (Ekramipooya et al., 2023). On the other hand, Term Frequency-inverse Document Frequency (TF-IDF), Latent Dirichlet Allocation (LDA) and Naïve Bayes were applied to parameters, guidewords, causes and consequences, using 5503 datapoints, to predict likelihood, severity, and risk levels, achieving accuracies of 83.2%, 84.5%, and 85.7%, respectively (Wang and Gu, 2022). These studies utilized different algorithms, specific textual input data (HAZOP report elements), varying text lengths, different target variables, and varying sample sizes, yielding different results. Furthermore, they overlooked the mining of all HAZOP reports' elements, including node description, guide words, parameters, deviations, causes, consequences, safeguards, and recommendations. Addi-

tionally, BOW and TF-IDF algorithms have limitations, including the difficulty of capturing polysemy, sparse representation, and the ignorance of semantic relationships between words, which can result in less accurate text representations. To address these issues, this paper aims to: (1) introduce the Global Vectors for Word Representation (GloVe) algorithm, a robust word embedding technique widely used in various NLP tasks which has not been applied in HAZOP studies; (2) apply sentence bidirectional encoder representation from transformers (SBERT) embeddings to HAZOP data for rich, context-aware representations; (3) implement zero-shot natural language inference (NLI) pipeline to predict labels without any task-specific training; (4) compare the performance of TF-IDF, TF-IDF and LDA, GloVe, and SBERT representations, each paired with ML classifiers, alongside the zero-shot model against the BOW algorithm using the available data from the Tankage Area case study; and (5) evaluate their effectiveness in accelerating hazard identification across all HAZOP components and enhancing safety in infrastructure operations.

The findings contribute to advancing the automation of HAZOP study in infrastructure projects, offering a framework for selecting the most suitable NLP strategy and ML classifier for specific applications. Automating historical HAZOP reports in different areas using these advanced technologies like ML and NLP increasingly supports less experienced engineers and participants in efficiently conducting the HAZOP process, ensuring continuity and knowledge retention.

The case studies in this research were selected for several key reasons. The first case study was chosen due to the rarity of publicly available HAZOP data, as such datasets are seldom found in the literature. This case study raised important questions by focusing solely on predicting deviations based on causes, ignoring other HAZOP elements. While BOW technique achieved a high accuracy of 92%, it prompted further exploration of how other NLP algorithms would perform and how BOW would fare when applied to a comprehensive set of HAZOP elements, such as node name, guide words, parameters, causes, consequences, safeguards, and recommendations. The second case study used a traditional HAZOP approach for hoisting operations in prefabricated buildings, which took more than 20 days, and demonstrated the need for automating HAZOP studies in construction operations to reduce reliance on experts and shorten the time required. The study's published data provided a basis for testing different algorithms on all HAZOP elements to assess their effectiveness in this context.

The remainder of this paper is organized as follows: Section 2 reviews literature studies on applying NLP in HAZOP studies. Section 3 outlines the methodology. Section 4 presents a case study conducted in a chemical plant and its findings. Section 5 showcases a case study focused on automating HAZOP reports for hoisting operations in prefabricated buildings. Section 6 discusses the results, and Section 7 concludes the study.

2. Literature Review

Natural Language Processing (NLP) enables computers to process, understand, interpret, and manipulate human language meaningfully and effectively (Manning et al., 2014). This technology is essential for making HAZOP study reports accessible to machine learning (ML) algorithms, which cannot directly process text. NLP techniques can extract valuable insights from these reports for further analysis. NLP encompasses word embedding algorithms for knowledge representation and topic modeling techniques for clustering and topic representation. Advanced text modeling algorithms such as Word2Vec, which uses neural networks to generate word embeddings (Mikolov et al., 2013), Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2018), Embeddings from Language Models (ELMo) (Peters et al., 2018), and Generative Pre-trained Transformers (GPT) are well-suited for analyzing large datasets. Conversely, conventional methods like Bag of Words (BOW), Term Frequency-Inverse Document Frequency (TF-IDF), and Global Vectors for Word Representation (GloVe) are better suited for smaller datasets (Ekramipooya et al., 2023). Topic modeling algorithms play a crucial role in clustering sentences and documents. Latent Dirichlet Allocation (LDA) is widely recognized for its efficiency, speed, and effectiveness in clustering large-scale data (Suh, 2021). More recently, Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) has gained popularity for clustering and text analysis due to its robustness and ability to handle complex datasets (Grootendorst, 2022).

The current literature employed various NLP algorithms to convert textual data extracted from HAZOP reports into numerical representations that can then be analyzed by ML algorithms. BERT model was trained on Chinese text reports to assist smaller chemical plants in performing security analysis and enhancing safe production (Feng et al., 2021). The model utilized BERT as a word embedding layer, leveraging bidirectional transformers for training, and incorporated bidirectional long short-term memory (BiLSTM) with an attention mechanism for feature extraction. The model classified consequence severity in HAZOP studies using 2,075 consequence description datapoints as input features. It achieved precision, recall, and F1-scores of 88.6%, 88.2%, and 88.2%, respectively, outperforming the Word2Vec + BiLSTM + Attention model, which scored 85.8%, 84.8%, and 85.1% (Mikolov et al., 2013). While the results are primarily applicable to Chinese, the model can be adapted for English or other languages by replacing the language-specific BERT model and training corpus (Feng et al., 2021). An NLP framework was developed, combining rule-based phrase matching and a spaCy Named Entity Recognition (NER) model to extract causes, consequences, and hazards from free-text occurrence reports for validating an aircraft sub-system HAZOP study (Ricketts et al., 2022). The NER model demonstrated strong performance in predicting consequences, achieving precision, recall, and F1-scores of 85.71%, 85.47%, and 85.59%, respectively. This success was attributed to the tendency of reporters to clearly describe consequences, as effects are easier to articulate. In contrast, the prediction of causes showed lower metrics, with precision, recall, and F1-scores of 66.2%, 38.84%, and 48.96%, respectively. This was due to ambiguous language (e.g., "it was suspected..."), incomplete occurrence descriptions, and events where no fault was identified. Such challenges limited the accurate identification of causes in the dataset. A deep learning model (ELMo-DCNN-BiLSTM-CRF) was proposed for performing named entity recognition (NER) on Chinese HAZOP documents in the petrochemical industry, focusing on identifying material and equipment terms in HAZOP analyses (Peng et al., 2021). The dataset comprises 102 nodes and 5,000 effective sentences, derived from HAZOP reports from indirect coal liquefaction project. The preprocessed data were input into Embeddings from Language Models (ELMo) and a double convolutional neural network (DCNN) to extract rich character features. A bidirectional long short-term memory (BiLSTM) network captured long-distance semantic information, and results were decoded using a conditional random field (CRF) before output. The results revealed higher identification accuracy for equipment compared to material, with precision rates of 92.59% and 86.08%, recall rates of 93.52% and 83.95%, and F1-scores of 93.05% and 85.00%, respectively. A deep learning with multifractal (DLF) model was introduced for hazard event (HaE) classification, leveraging the time-series nature of HaE data (Wang et al., 2023). The study utilized HAZOP reports from 18 processes as cases, compiling a dataset of 5,869 labeled HaEs for classification based on severity, possibility, and risk. The methodology began with vectorizing HaEs into time-series data using BERT. A hazard multifractal detrended fluctuation analysis (HmF-DFA) was then applied to generate HaE fractal series by analyzing the time-series data. Finally, a hierarchical gating neural network (HGNN) was developed to process these fractal series and classify HaEs. The model achieved the best results for the severity aspect, with precision, recall, and F1-score of 83.04%, 85.99%, and 84.09%, respectively, compared to 70.45%, 73.29%, and 71.51% for the possibility aspect, and 73.07%, 79.02%, and 74.77% for the risk aspect. Table 1 presents the key characteristics of the various NLP models discussed in the literature.

Table 1
Characteristics of various literature NLP models used in HAZOP studies.

Method	Features	Entries	Label	Results	Industry	Reference
BERT + BiLSTM +Attention	Consequence description	2075	Severity	Precision 88.6%.	Chemical	(Feng et al., 2021)
TF-IDF + LDA +GNB	Parameters, Guidewords, causes and consequences	5503	Likelihood	Accuracy 83.2%	Petro-chemical	(Wang and Gu, 2022)
			Severity	Accuracy 84.5%		
			Risk Level	Accuracy 85.7%		
BOW+ML classifiers	Causes	42	deviations	Accuracy 92%	Chemical	(Ekramipooya et al., 2023)
Rule-based with a spaCy NER	Free-text occurrence reports	437 reports	Causes	Precision 85.71%	Aircraft sub-system	(Ricketts et al., 2022)
			consequences	Precision 66.2%		
ELMo-DCNN-BiLSTM-CRF	Chinese HAZOP reports	5000 sentences	Equipment and Material classification	Eq*. Precision 92.59%	Petro-chemical	(Peng et al., 2021)
				Eq. Recall 93.52%		
				Eq. F1-Score 93.05%		
				Mat*. Precision 86.08%		
				Mat. Recall 83.95%		
				Mat. F1-Score 85.00%		
BERT + HmF-DFA + HGNN	Process node, Deviation, Consequences, Causes, Measures, and Suggestions	5869 hazard events (HaE)	Prediction of possibility, severity, and risk level	Sev*. Precision 83.04%	Petro-chemical	(Wang et al., 2023)
				Sev. Recall 85.99%		
				Sev. F1-Score 84.09%		
				Pos*. Precision 70.45%		
				Pos. Recall 73.29%		
				Pos. F1-Score 71.51%		
				Risk. Precision 73.07%		
				Risk. Recall 79.02%		
				Risk. F1-Score 74.77%		
BERT + UMAP + HDB-SCAN + MLP	Causes	196	Prediction of Recommendations	Accuracy 93.7%	Petro-chemical	(Ekramipooya et al., 2024)
	Consequences	196		Accuracy 89.5%		

* Eq.: Equipment, Mat.: Material, Sev.: Severity, and Pos.: Possibility.

Table 2
the characteristics of the two case studies examined in this study.

Case	Nodes	Features	Datapoints	Label	Industry
Case study (1)	3	Causes	42	deviation	Chemical
Case study (2)	6	Node description, Parameters, Guidewords, deviations, causes and countermeasures	23	Consequence	Construction

The aforementioned studies demonstrate the use of different NLP algorithms, diverse datasets and the resulting outcomes, influenced by factors such as feature definition, text length, target variables, number of labels, and the size of datasets used for analysis. For instance, Table 1 illustrates the variation in textual input lengths, ranging from consequence descriptions in the BERT + BiLSTM + Attention model to combinations of parameters, guidewords, causes, and consequences in the TF-IDF + LDA + GNB model, and further extending to process nodes, deviations, consequences, causes, measures, and suggestions in the BERT + HmF-DFA + HGNN model. Additionally, the number of entries (datapoints) varies across the models, impacting the scale of the dataset employed for training and evaluation. Furthermore, the results indicate a broad spectrum, ranging from 66.2% to 93.52%, across classification models and various target variables. These considerations are paramount, as they can significantly influence the performance of the models. Given the disparities observed in these findings, our research aims to conduct a comparative analysis among traditional NLP algorithms under limited datasets including BOW, TF-IDF, TF-IDF + LDA and GloVe, and sentence embedding such as sentence-BERT (SBERT) to identify the most suitable NLP strategy in automating HAZOP reports within infrastructure projects. Additionally, a zero-shot classification model is implemented to assign a label to a text without prior training.

Given the non-utilization of NLP in HAZOP studies within subareas of infrastructure projects and the most existing literature studies were in the chemical and petrochemical plants, the practical implementation initially applied NLP strategies in the chemical industry. Subsequently, these algorithms were extended to the hoisting operations of prefabricated buildings for a comparative analysis of the results. It is crucial to note that our approach involves considering data from all HAZOP report elements in the analysis particularly emphasized in the second case study, ensures that our research aligns closely with real-world HAZOP applications, thereby enhancing the relevance and reliability of our comparative findings.

3. Methodology

The overall workflow of utilizing NLP algorithms (BOW, TF-IDF, TF-IDF + LDA, GloVe, and SBERT) and ML classifiers to automate HAZOP reports is depicted in Figure 2, detailing each step from preprocessing text data to obtaining the results and evaluation. Additionally, the workflow for implementing zero-shot classification model is described separately in Section 3.7.

3.1. Data Acquisition

The data was obtained based on conventional HAZOP study reports. The dataset utilized in this study comprises two distinct case studies. The first case involves the utilization of causes as a feature extracted from HAZOP report of the actual unit processes in a tankage area process equipment which was recently conducted in a chemical

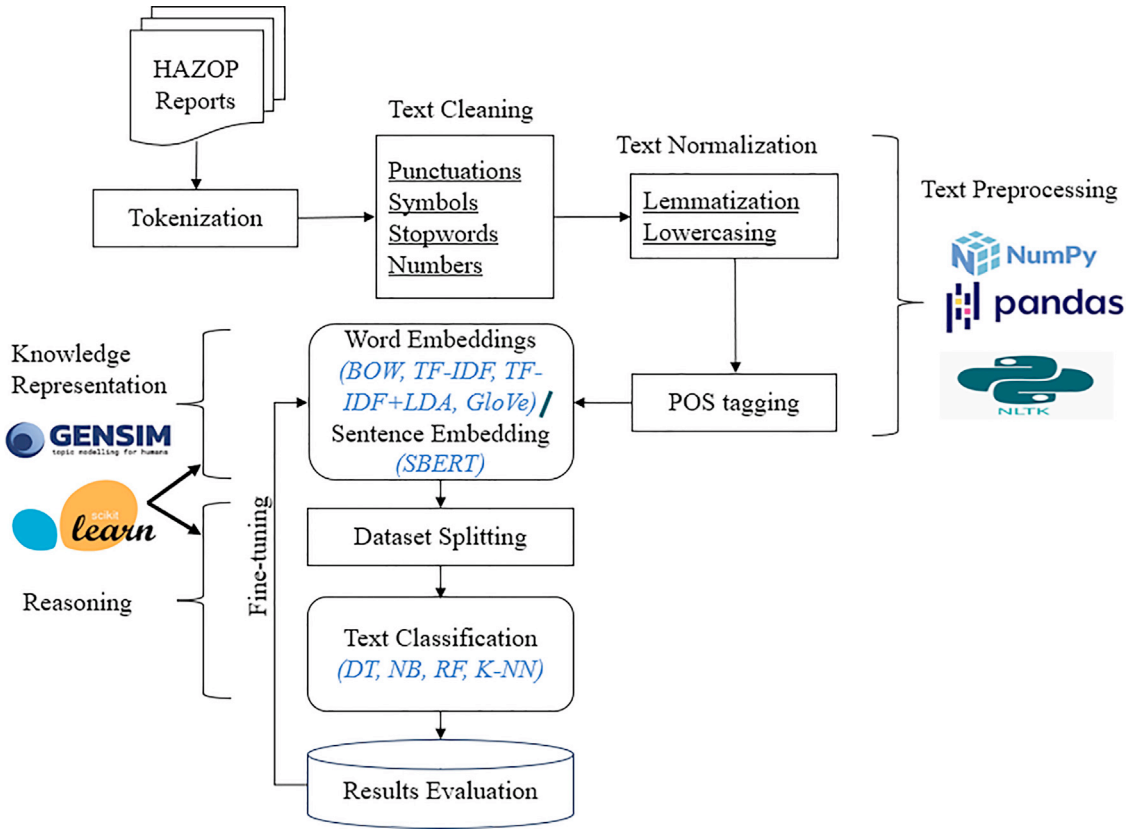


Figure 2. Overall workflow of NLP strategies to automate HAZOP reports.

plant in Iran. This dataset encompasses 3 nodes and 25 records and 42 records after oversampling and was employed to predict deviations (Ekramipooya et al., 2023). The second case study encompasses node description, parameters, guide words, deviations, causes, and countermeasures as features. This dataset was utilized to predict the impact of unsafe behaviors of workers (consequence) during hoisting operations of concrete prefabricated buildings in China to mitigate unsafe behaviors, comprising 6 phases (nodes) and 23 datapoints (Zhu et al., 2022). Table 2 illustrates the details of the two case studies utilized in this research.

3.2. Data Preprocessing

Text data preprocessing is a fundamental step, aimed at converting raw textual data into a clean and structured format suitable for analysis. In this research, the Natural Language Toolkit (NLTK) package (Bird, 2006) was utilized to carry out the following text preprocessing steps:

- **Tokenization:** Tokenization is the process of breaking text into smaller components, or tokens, based on meaning rather than just spaces. Tokens typically represent individual words or meaningful units within a text string (Liu et al., 2021). For instance, the sentence “Excess pump is brought in service by error” is tokenized into eight tokens: ["Excess", “pump”, “is”, “brought”, “in”, “service”, “by”, “error”];
- **Text Cleaning:** This preprocessing task focuses on removing formatting from text, including punctuation, extra spaces, numbers, and stop words. The purpose is to eliminate words that add little to no value or may interfere with text analysis. Stop words, such as “an” and “the,” are extremely common words that provide minimal analytical value. In this work, the stop words list from NLTK corpus was utilized (Bird, 2006);
- **Text Normalization:** This step involves various preprocessing tasks, including stemming, lemmatization, upper or lowercasing. In this work, both lowercasing and lemmatization were used. Lemmatization reduces words to their root forms and is more accurate than stemming, as it considers the word’s meaning and context. While stemming is faster, it sacrifices linguistic accuracy by not considering context (Toman et al., 2006). This work prioritizes accuracy over speed, thus opting for lemmatization over stemming (Liu et al., 2021);
- **Part-of-speech (POS) tagging:** POS tagging assigns a part of speech to each word in a text string, such as noun, verb, or adjective. This process enhances lemmatization accuracy by considering the grammatical role of words, which is crucial when a word has multiple meanings. ‘Pos_tag’ function is used to determine and assign these tags, thereby improving the overall text analysis (Hai et al., 2023).

The text preprocessing steps implemented in this study to prepare the dataset involved multiple stages to ensure the data was clean, standardized, and suitable for analysis. Initially, special characters and symbols were removed to retain only meaningful content. The text was then converted to lowercase to standardize the data and minimize variations caused by case sensitivity. Tokenization was applied to break the text into smaller components, or tokens, facilitating further processing. Stop words were removed to focus on the most relevant information. Lemmatization was used to normalize the text by transforming words into their base or dictionary forms, ensuring consistency while preserving their contextual meaning. Finally, part-of-speech (POS) tagging was employed to categorize each word grammatically, identifying it as a noun, verb, adjective, or other parts of speech, thereby enriching the text with valuable contextual information for subsequent analysis.

Due to the small size of the dataset used in both case studies, the implementation and training of the models were carried out on Google Colab CPU. This platform was chosen for writing and running codes as the data processing tasks were not computationally intensive and could be efficiently performed without significant processing time on the CPU.

3.3. Knowledge Representation (Word Embedding)

After preprocessing, the text of each datapoint or record is transformed into clean and normalized tokens. These tokens are then converted into numeric features to facilitate the application of machine learning and text mining techniques. As previously mentioned, numerous word embedding models exist for training word vectors in natural language processing including BOW, TF-IDF, Word2Vec, BERT, GloVe, LDA. Considerable attempts have been made to compare these algorithms for creating word embeddings on different datasets for different purposes. For example, (Toshevska et al., 2020) analyzed several word embedding methods including Word2Vec, FastText, LexVec and GloVe to evaluate their performance on capturing word similarities and found that GloVe and FastText outperformed the other methods. The literature on HAZOP analysis has commonly utilized BOW, TF-IDF, and LDA in classification models to automate HAZOP reports. Therefore, we adopted these strategies in our study. Additionally, we incorporated the GloVe method because it addresses the limitations of other strategies by utilizing a word-word co-occurrence matrix of words (Gerth, 2021).

3.3.1. Bag of Word (BOW)

BOW is a text representation method that captures the frequency of word occurrences within a document. It creates a “bag” of unique words from the text while disregarding grammar, syntax, and word order (Qader et al., 2019). This simplified representation focuses solely on the presence or absence of words and their frequency, making it a foundational technique for converting text into a numerical format suitable for machine learning and text analysis tasks (Kim et al., 2017).

3.3.2. Term frequency-inverse document frequency (TF-IDF)

A challenge with scoring word frequency in the Bag of Words (BOW) approach is that common words can dominate the representation despite offering little “informational content,” while rarer, domain-specific terms, which are more meaningful, may be overshadowed. Term frequency-inverse document frequency (TF-IDF) technique addresses this by weighting words based on their importance across the dataset. TF-IDF calculates the relative frequency of a word in a specific document, adjusted by its inverse frequency across all documents, resulting in a term-document matrix of TF-IDF scores. This approach assigns higher weights to words that occur frequently in specific documents but are rare across the dataset. The formula for calculating the TF-IDF value for a term t in a document d within the term-document matrix is as follows (Kim et al., 2019):

$$TF - IDF(t,d) = TF(t,d) * IDF(t) = TF(t,d) * \log, \quad (1)$$

Where:

- TF (t, d): The frequency of term (or word) t in document (record) d,
- N: The total number of documents in the dataset,
- df(t): The document frequency, representing the number of documents containing the term t.

To account for cases where a term appears in every document, resulting in an IDF(t) of zero, a value of one is added inside the logarithmic term during IDF(t) calculation. The equation above computes the TF-IDF score for each term t in every document d, resulting in a TF-IDF matrix with dimensions N×M, where M is the total number of unique words in the dataset. This matrix serves as the input dataset for prediction and classification algorithms.

3.3.3. Global Vector for word representation (GloVe)

GloVe is an unsupervised learning algorithm developed at Stanford by (Pennington et al., 2014) for generating vector representation or embeddings of words. It leverages global word-word co-occurrence statis-

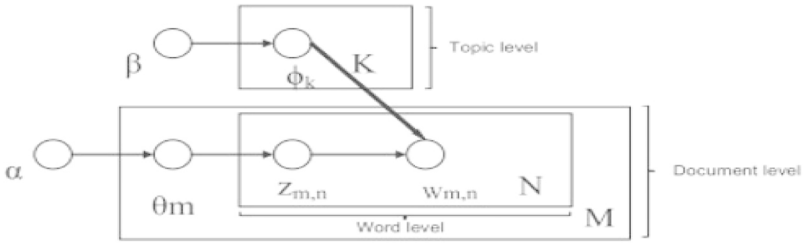


Figure 3. The Latent Dirichlet Allocation (LDA) concept. Source: (Toman et al., 2006).

tics from a corpus. Its primary goal is to produce dense vector representations of words that capture their semantic relationships. GloVe combines the strengths of local context window methods, such as the skip-gram model, with global matrix factorization techniques to create low-dimensional word representations. While the skip-gram model excels at analogy tasks, it does not fully utilize corpus statistics because it lacks training on global co-occurrence counts. In contrast, GloVe focuses on global context by using a word-word co-occurrence matrix, which provides a broader perspective compared to Word2Vec, which primarily relies on local context. This global focus allows GloVe to offer advantages in certain applications by capturing deeper semantic relationships between words. The algorithm achieves this through the use of a co-occurrence matrix, as represented by the following equation:

$$P(j|i) = \Pr obability\ (jisinthecontextof\ i) = \left(\frac{Xij}{Xi}\right)_a, \quad (2)$$

Where,

- Xij = Co-occurrence Matrix[i][j]
- $Xi = \sum_{1}^{k} Xik$ (The total number of times any word appears in the context of word i)
- α: a hyperparameter that balances rare and frequent co-occurrences.

3.3.4. Topic Modeling: Latent Dirichlet allocation (LDA)

Latent Dirichlet Allocation (LDA) is a probabilistic generative model designed to uncover the underlying thematic structure in a collection of documents. It assumes that each document is a mixture of multiple topics, and each topic is a distribution over words (Niu et al., 2019). By learning these distributions, LDA identifies the hidden topics that characterize the dataset. The model assigns probabilities to words and documents, making it possible to determine how strongly a word is associated with a topic and how topics are distributed across documents. Its ability to reveal hidden patterns in unstructured text data has made it one of the most widely used and fundamental approaches in topic modeling (Blei et al., 2003). Although LDA is an unsupervised learning technique that does not utilize label information, (Wang et al., 2012) attempted to enhance document classification performance by incorporating label information alongside LDA-based document representation.

To identify the optimal number of topics or clusters (K) for an LDA model, perplexity is commonly evaluated across different topic models (Huang et al., 2017). Perplexity quantifies how well a probability model predicts a dataset, with lower perplexity scores indicating better model performance. The formula for calculating perplexity is:

$$P = exp\left(-\frac{\sum_{m}\sum_{i}^{Nm} lnp(Wm,i)}{\sum_{m} Nm}\right), \quad (3)$$

Where:

Nm represents the word frequency in document m.
Wm,i denotes the i-th word in document m.

In LDA, each document is modeled as a multinomial distribution over latent topics, and each topic is represented as a multinomial distribution over words. The generative process for LDA assumes that documents in a collection are created through the following steps, as illustrated in Figure 3:

1. For each document m in the collection M, sample topic proportions θm from Dirichlet distribution Dir(α);
2. For each word placeholder n in document m:
Select a topic $z_{m,n}$ randomly based on the topic proportions θ_m;
Select a word $w_{m,n}$ randomly from the multinomial distribution ϕ_k associated with the chosen topic $z_{m,n}$.

In the process described above, the parameters α and β are vectors of hyperparameters that define the Dirichlet priors. Specifically, α sets the prior on θ, which represents the distribution of topics across all documents, while β sets the prior on ϕ, which represents the word distribution for each topic.

In this work, the LDA algorithm was combined with TF-IDF to enhance text embeddings by leveraging both term frequency and topic modeling. This combination creates a feature vector that captures individual word importance (TF-IDF) and overall topic distribution (LDA), providing a richer and more nuanced representation of the text. This enriched feature representation can improve prediction model performance by offering a comprehensive understanding of the text data, reducing dimensionality, and incorporating context and semantics into the feature vectors.

3.4. Knowledge Representation (Sentence Embedding)

Traditional word-level embeddings (e.g. GloVe, Word2Vec) produce static vectors that fail to capture full sentence semantics, motivating the use of transformer models like BERT for sentence representation. A common approach is simply to average the final BERT token embeddings or to use the special [CLS] token output as a sentence vector, but this common practice yields rather bad sentence embeddings, often worse than averaging GloVe embeddings (Reimers and Gurevych, 2019). To address this, Sentence-BERT (SBERT) was proposed as a modification of BERT (Bidirectional Encoder Representations from Transformers) specifically designed for generating dense, fixed-size sentence embeddings that capture semantic meaning efficiently. Each input sentence is tokenized and mapped to token, segment, and positional embeddings, summed and passed through BERT's bidirectional encoder. SBERT then applies mean pooling over the final-layer hidden states to yield a single 768-dimensional vector per sentence (Aytaç and Khayet, 2023). Table 3 illustrates the main differences between traditional NLP algorithms and sentence embedding algorithm

After text preprocessing and splitting the dataset into training and testing sets, we proceeded to implement NLP algorithms to convert the text into numeric features utilizing scikit-learn and Gensim libraries. Firstly, we implemented BOW using scikit-learn's CountVectorizer, which represents each document or record as a vector of word frequencies. Next, we utilized TF-IDF with scikit-learn's TfidfVectorizer to weigh word importance based on their frequency within a document and across the entire corpus. Additionally, we experimented with TF-IDF combined with LDA using scikit-learn's LatentDirichletAllocation, which incorporates topic modeling to further enhance feature representation. Moreover, we leveraged the GloVe method with Gensim to capture

Table 3
Key Differences Among NLP Algorithms.

Aspect	BOW	TF-IDF	GloVe	SBERT
Representation Type	Represents features based on word frequency counts (Qader et al., 2019).	Assigns higher weights to terms important in a datapoint relative to their occurrence across the corpus(Kim et al., 2019).	Generates dense word embeddings based on global word co-occurrence statistics (Pennington et al., 2014)	Creates dense embeddings that capture sentence-level semantics using transformer-based architectures (Reimers and Gurevych, 2019)
Semantic Understanding	None	None	Word-level semantics	Sentence-level semantic context
Contextual Meaning	No context	No context	Limited (word co-occurrence) (Reimers and Gurevych, 2019)	High (context-aware representations) (Aytaç and Khayet, 2023)
Dimensionality	sparse, high-dimensional representations	sparse, high-dimensional representations	Low-dimensional, dense	Low-dimensional, dense
Computational Efficiency	High (simple calculations)	High (efficient calculations)	Moderate (training required)	Lower (requires GPU, high resources)

semantic relationships between words by training word embeddings based on co-occurrence statistics in the corpus. Finally, we employed the Sentence-BERT (SBERT) model using the sentence-transformers library, specifically the all-mpnet-base-v2 model, to generate sentence-level embeddings to capture semantic meaning across sentences and datapoints.

To enhance the performance of text classification models through different feature representations, we fine-tuned our methodology by exploring TF-IDF+LDA with varying LDA component numbers, ranging from 2 to 10, to discern the optimal configuration for feature representation and the decision to employ 2 components reflected an optimal level of abstraction based on perplexity score. Simultaneously, The GloVe model was implemented with varying window sizes, ranging from 3 to 10, to systematically examine its impact on the models' overall performance. The choice of a window size of 5 captured the nuanced semantic relationships between words within a moderate range. The vector size was set to 100, defining the dimensionality of the word embeddings used in the model. To limit the influence of very frequent co-occurrences, Xi (in Eq. 2) was set to 100, ensuring that the weighting function assigns higher weights to word pairs that occur frequently, while reducing the impact of very frequent pairs, such as common stopwords ("the," "and"), which could skew the model. The smoothing parameter α was set to 1, allowing the weighting function to decrease the influence of very frequent co-occurrences without completely disregarding them. These parameters were consistently applied across both case studies to evaluate their effectiveness.

3.5. Reasoning

In this study, Naïve Bayes (NB), Decision Tree (DT), K-Nearest Neighbors (KNN), and Random Forest (RF) classifiers were adopted as base models for text classification. Each classifier offers distinct advantages in addressing text classification tasks. Naïve Bayes, grounded in Bayes' theorem, assumes feature independence, making it computationally efficient and straightforward to implement. Decision trees use a hierarchical structure to recursively partition the feature space according to attribute values, resulting in interpretable models that can capture complex decision boundaries. K-Nearest Neighbors classify data points by considering the majority of their nearest neighbors in the feature space, effectively handling nonlinear decision boundaries and varying local densities. Random Forest, an ensemble of decision trees, enhances prediction accuracy by aggregating the outputs of multiple trees to reduce overfitting and improve robustness. These classifiers are well-suited for text classification tasks (Bouguelia et al., 2013; Hai et al., 2023) due to their computational efficiency and ease of implementation in practical systems. Moreover, HAZOP studies have applied these algorithms for reasoning and high accuracy predictions because of their diversity which benefit from their diversity and suitability for various applications (Ekramipooya et al., 2023). Therefore, we utilized these algorithms to forecast class labels and identify the best NLP strategy model processing.

3.6. Performance Assessment

We used four performance metrics—accuracy, precision, recall, and F1-score—to evaluate the effectiveness and quality of the models. Assessing classifier performance is crucial for understanding its ability to differentiate between classes, and a confusion matrix serves as an essential tool for this purpose. The confusion matrix provides insights into correct classifications through true positives (TP) and true negatives (TN), while also identifying errors through false positives (FP) and false negatives (FN). The evaluation metrics used in this study are calculated as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{4}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

$$\text{F1} - \text{Score} = \frac{2 * Precision * Recall}{Precision + Recall} \tag{7}$$

Although the scoring criteria discussed thus far primarily pertain to binary classification models, the one-versus-all strategy implemented in scikit-learn extends these metrics to multiclass scenarios by using appropriate averaging methods (Raschka and Mirjalili, 2019). This study employed the train-test split approach to evaluate the machine learning classifiers. The dataset was divided into a training set, used to fit the algorithms, and a testing set, used to measure their performance. This methodology provides a robust framework for comprehensively assessing the efficacy of the classifiers under consideration.

3.7. Zero-shot Text Classification

Zero-shot text classification (Yin et al., 2019) assigns each input directly to a target category without any task-specific training by treating the input as a "premise" and each label as a "hypothesis," then using a pre-trained natural language inference (NLI) model to score how strongly the premise entails each hypothesis. This eliminates the need to hold out labeled examples for training, allowing immediate categorization of HAZOP data. We use Hugging Face's facebook/bart-large-mnli—a BART-large transformer with over 400 M parameters (Yin et al., 2019)—which employs an encoder–decoder architecture with self-attention to produce context-aware embeddings. The attention mechanism captures the semantic relationships between different words taking into account also the position of the sentence. The architecture of transformer-based models is shown in Figure 4 and is based on the groundbreaking work outlining the initial proposal of transformer models (Vaswani et al., 2017). For every datapoint (the "premise"), we call the Hugging Face's NLI pipeline with our set of hypotheses and receive entailment scores. Different thresholds were

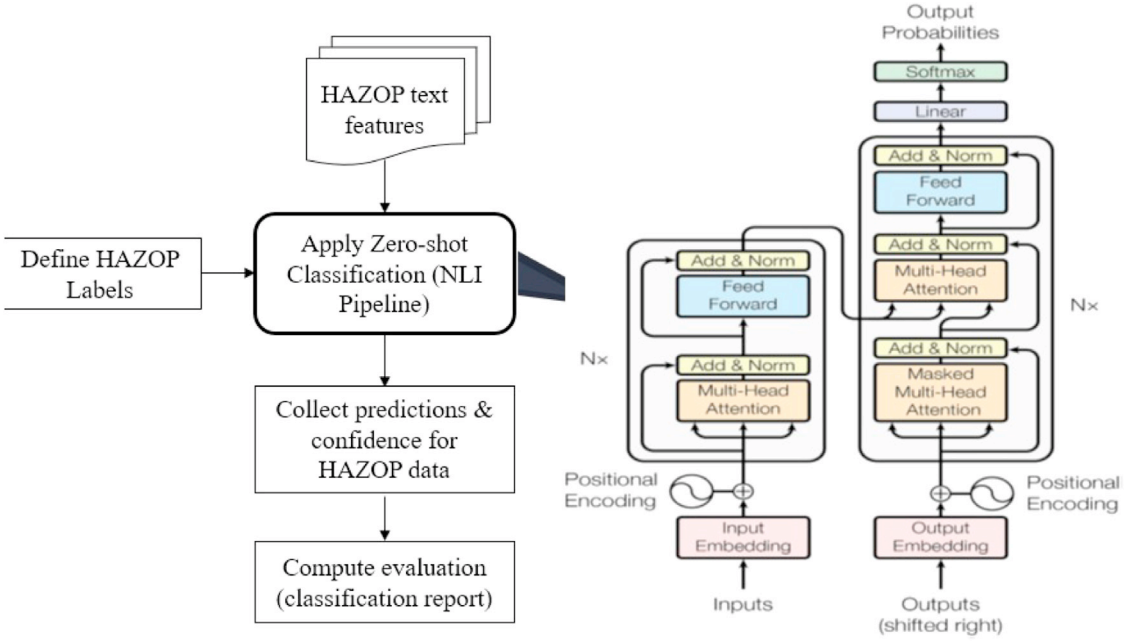


Figure 4. High level figure using zero-shot classification; Transformer model architecture (Vaswani et al., 2017).

tested to select the label whose hypothesis has the highest entailment probability and convert it back to the original code. The resulting predictions and confidence scores are appended to the dataset and evaluated—against true labels—using weighted precision, recall, and F1-score via sklearn.metrics.classification_report. A high-level diagram of the zero-shot classification is presented in Figure 4.

4. Case Study (1)

Building on the aforementioned methodology, this research applied text data mining algorithms to two case studies for intelligent hazard prediction and automatic HAZOP analysis. The first case study examines an actual unit in the chemical industry. Ekramipooya et al., (2023) applied bag of words (BOW) in Tankage Area process equipment consisting of 3 nodes including Methanol Storage Tanks (TK-0101A/B), Pump Station (P-0101 A/B/C and P-0103 A/B/C), Pig Launcher (PL-0101), and Slop Drum (D-0101), as illustrated in Figure 5.

The Bag of Words (BOW) algorithm was employed to convert causes into feature vectors for predicting deviations. The dataset consisted of 6 labels, with label frequencies ranging from 1 for “low pressure” to 7 for “no or less flow,” indicating an imbalanced dataset. To mitigate the challenges posed by the dataset’s imbalance and limited size, random oversampling was applied, resulting in 42 tuples post-oversampling. The data, encompassing causes and deviations, was extracted fro m a conventional HAZOP study report. Table 4 presents examples from each node of this report. According to Table 4, abbreviations used in this case study are NLF, MF, HP, LP, HL, and LL, representing no or less flow, more flow, high pressure, low pressure, high level, and low level respectively.

For the modeling process, 67% of the data (28 tuples) was used for training, and the remaining 33% (14 tuples) constituted the test set. Classifier performance in predicting class labels was evaluated using Accuracy and F1-score as metrics. We used the data of this case study to apply other NLP algorithms (TF-IDF, TF-IDF + LDA, GloVe, and SBERT) to analyze the performance of processing textual data and apply ML classifiers to predict deviations. Additionally, zero-shot NLI pipeline (facebook/bart-large-mnli) is applied to predict deviations without training data.

4.1. Results

Based on the accuracy results provided for different NLP methods and machine learning classifiers compared to BOW values in Table 5, it is obvious that BOW consistently delivers high accuracy across classifiers. TF-IDF demonstrates strong performance, particularly with DT model (86%), indicating its versatility and effectiveness in capturing important features for the given task. However, it exhibits lower performance across other models. GloVe embeddings show competitive performance compared to TF-IDF and TF-IDF + LDA methods and achieve the highest accuracy with GNB (89%), but the results vary across models. The deviations in this case study are more evenly distributed and balanced. Therefore, GNB can perform better. GloVe’s smooth, continuous feature distribution aligns well with GNB’s assumption of normally distributed data, enabling more accurate probability estimations. SBERT achieved poor performance because of short cause descriptions and limited number of samples for each label (5–8 samples per class), hindering the model’s ability to learn from training data.

Table 4
HAZOP report Samples, including causes and deviations (Ekramipooya et al., 2023).

Node	Causes	deviations
1	Decreased/ cut-off flow from upstream due to any reason	NLF
1	Increased flow from upstream due to any reason	MF
2	Closure or not opening of manual valves around Pig Launcher	NLF
2	Heat source such as sun radiation during box-up condition	HP
3	Closure of PCV by failure or error when it should be open	LP
3	Drainage to drum and pump does not start in proper time due to any reason	HL
3	Pump does not stop in proper time due to any reason	LL

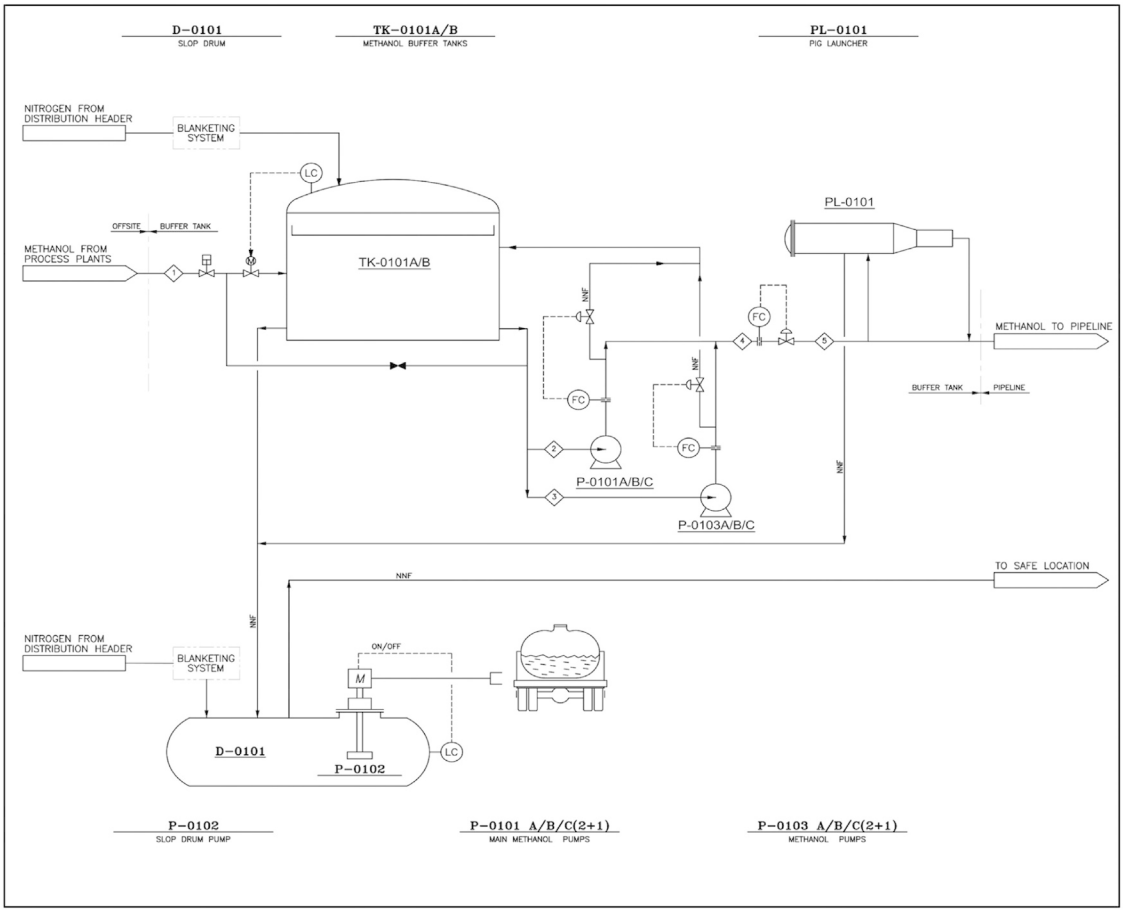


Figure 5. The Process Flow Diagram of case study (1), consisting of three nodes (Ekramipooya et al., 2023).

Table 5
NLP models' Accuracy of different ML classifiers compared to BOW (case study 1).

Model	DT	RF	GNB	KNN	Reference
BOW+ML classifiers	0.92	0.85	0.85	0.78	(Ekramipooya et al., 2023)
TF-IDF+ML classifiers case (1)	0.86	0.79	0.71	0.36	The Authors
TF-IDF+ LDA +ML classifiers case (1)	0.57	0.57	0.64	0.43	The Authors
GloVe + ML classifiers case (1)	0.78	0.78	0.89	0.67	The Authors
SBERT + ML classifiers case (1)	0.43	0.57	0.43	0.57	The Authors

Following the evaluation based on accuracy and because the data was balanced, we extended our assessment to include the F1-score, which provides a more detailed measure of model performance by balancing precision and recall. The balanced nature of the dataset ensures that the models are not biased toward any specific class, allowing for a more comprehensive assessment of their capabilities. Figure 6 shows that the DT model exhibits high F1-scores (91%) with BOW features. Gaussian Naive Bayes (GNB) performs well, particularly with BOW (86%) and GloVe (85%), highlighting its ability to balance precision and recall effectively. However, the TF-IDF+LDA representation consistently yields lower F1-scores across models, suggesting that the additional LDA features might not contribute positively to the overall performance. GloVe embeddings showcase competitive F1-scores, with the models demonstrating a balanced capability in capturing both precision and recall. The balanced nature of the dataset enables a robust evaluation, shedding light on the models' effectiveness in discerning patterns and making accurate predictions across various feature representations.

Based on the findings in Table 5 and Figure 6, it is notable that K-nearest neighbors (KNN) exhibited the lowest performance among the classifiers in terms of predictive accuracy and F1-Score. This outcome may be attributed to the limited dataset, which might not adequately represent the true distribution of the data. KNN relies on the similarity of data points for making predictions, and a constrained number of data points may fail to capture the full complexity of the underlying distribution. Moreover, TF-IDF and BOW create high-dimensional and sparse feature spaces. KNN relies on distance metrics to find the nearest neighbors, and in high-dimensional spaces, the distance between data points becomes less meaningful. While GloVe captures semantic relationships between words, KNN may still struggle to leverage this information effectively in a sparse or high-dimensional feature space.

The zero-shot classifier based on facebook/bart-large-mnli achieved an overall accuracy of 0.52, precision of 0.46, recall of 0.52 and F1-score of 0.38. It performed well on "More Flow" and "No or Low Flow" (F1-scores of 0.64 and 0.70, respectively) but failed entirely to identify "High Level" and "Low Level" (all F1-scores = 0.00). This performance gap likely reflects the model's lack of exposure to HAZOP-specific terminology and the insufficiency of simple hypotheses ("high level," "high pressure," etc.) to capture subtle domain cues. Moreover, our evaluation set is small—42 examples total, with only 5–7 instances per under-represented class—so even a few misclassifications drive recall and F1-score to zero.

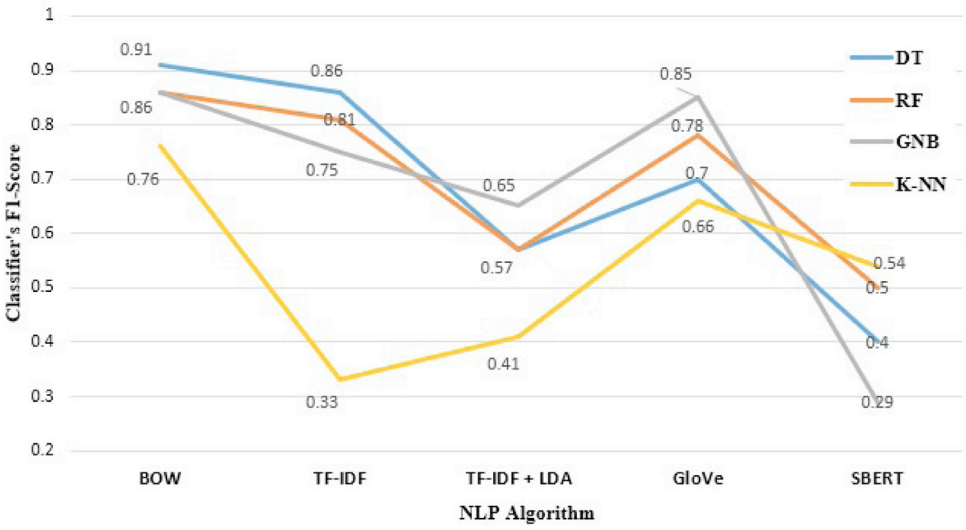


Figure 6. F1-Score of NLP models compared to BOW (case study 1).

Table 6
Examples of HAZOP results in hoisting operations of prefabricated buildings (Zhu et al., 2022).

Node	Lead Word	Parameter	Deviation	Reason	Countermeasure	Conseque-nce
loading and unloading phase	none	established workflow	The special iron frame for transportation is not fixed	The stability of the support system is poor	Regular inspection and maintenance of special transport frame	WH
stacking phase	other than	meet design requirement	The stiffness of the Member storage frame is poor	The stability of the support system is poor	Check and maintain component storage regularly	WH
lifting phase	Part of	the goal	The embedded lifting ring of the component falls off	The quality of assembled parts is poor	Improve and implement a comprehensive acceptance system for the entry of components	WI
installation phase	more	working hours or intensity	The lowering speed of components is too fast, or the components are not horizontal when falling	The level of safety awareness of construction personnel is low	Periodically carry out construction safety drills	WI
positioning phase	as well as	unnecessary actions	Prefabricated overlap plate-positioning correction squeezing hands	Poor safe operation training	Equip workers with protective equipment	WI
correction phase	Part of	the goal	Collapse of prefabricated composite plate	Stability of the support system is poor	Harden template connections	WD

5. Case Study (2)

This case study aims to explore utilization of NLP strategies for the automation of HAZOP reports in the operations of infrastructure projects, ultimately contributing to improved safety practices, reduced project timelines, and enhanced overall project outcomes in infrastructure projects. By leveraging advanced technologies such as NLP algorithms, infrastructure projects can benefit from a more systematic and data-driven approach to hazard identification. The dataset in this case study was obtained from conventional HAZOP results in concrete prefabricated buildings in China. Zhu et al., (2022) employed conventional HAZOP study to analyze and mitigate unsafe behaviors of construction workers during hoisting operations in prefabricated buildings. Initially, a questionnaire survey was conducted to collect data on safety risks and influencing factors at various stages of hoisting operations. Following this, HAZOP study was applied to identify deviations and changes of unsafe behaviors, along with their causes, consequences, and countermeasures. This process established a theoretical foundation for reducing unsafe behaviors by dividing hoisting operations into six phases aligned with the typical prefabricated construction sequence, which include component loading and unloading, stacking, lifting, installation, positioning, and correction. Table 6 details examples from each node of the 23 hazards identified in the literature study.

Building upon this groundwork, the present study utilized NLP algorithms, including BOW, TF-IDF, TF-IDF+LDA, GloVe, and SBERT on the same textual data. The zero-shot classification model was also implemented to assign consequence label without prior training. These approaches aim to automate the generation of HAZOP reports, providing construction engineers with a valuable tool for expedited hazard identification and enhanced safety measures in similar projects. The dataset encompasses node descriptions, parameters, guidewords, deviations, causes, and countermeasures as features, reflecting the multifaceted aspects involved in the hoisting operations. These features collectively provide a detailed representation of the operational context, deviations from expected behaviors, and potential causes for unsafe behaviors among construction workers. The target variable in this study is centered around consequences or the impact on workers' safety which WH, WI and WD stand for hurting workers, injures to workers, and worker death, respectively. By considering consequences as the target variable, the model aims to correlate the features with the potential adverse effects of unsafe behaviors during hoisting operations. The dataset under consideration comprises 23 data points, and the potential augmentation through oversampling would result in a modest increase to 27 datapoints, representing only four additional examples. As this provided no significant performance improvement and could introduce noise due to the short feature text length, oversampling technique was not pursued in this case study.

5.1. Results

The results of predictive modeling, as summarized in Table 7, show that the highest accuracy scores were observed with the GloVe embed-

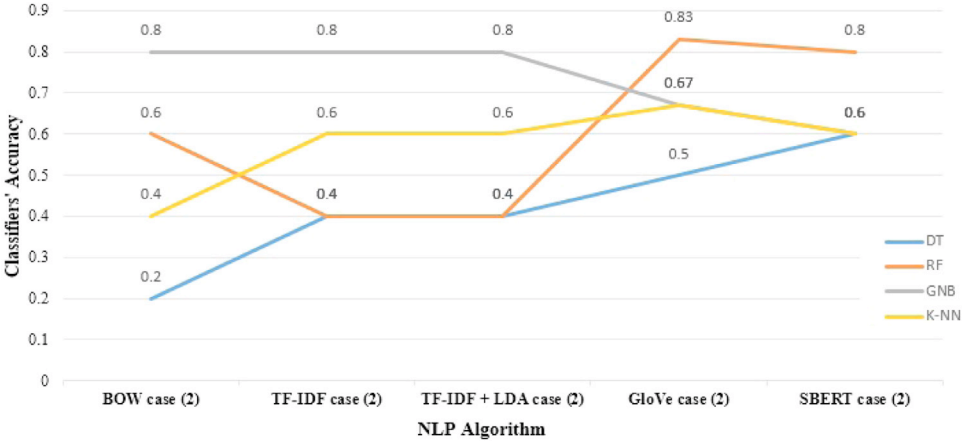


Figure 7. Predictive Accuracy in Hazard Prediction (Case Study 2).

Table 7
Accuracy results of NLP strategies (Case study 2).

Model	DT	RF	GNB	KNN
BOW+ML classifiers case (2)	0.2	0.6	0.8	0.4
TF-IDF+ML classifiers case (2)	0.4	0.4	0.8	0.6
TF-IDF+ LDA +ML classifiers case (2)	0.4	0.4	0.8	0.6
GloVe + ML classifiers case (2)	0.5	0.83	0.67	0.67
SBERT + ML classifiers case (2)	0.6	0.8	0.6	0.6

dings, particularly notable for Random Forests with an accuracy of 83%. This underscores the efficacy of semantic word representations in capturing intricate relationships within the dataset. Gaussian Naive Bayes (GNB) also performed consistently well, reaching 80% accuracy with both BOW and TF-IDF, indicating its robustness across different feature types. In contrast, the performance of BOW and TF-IDF declined significantly compared to the first case study, particularly with DT and RF, due to their limitations in modeling context and their susceptibility to high-dimensional sparsity—issues that become more pronounced with longer, more detailed input texts. The incorporation of LDA into TF-IDF did not notably improve accuracy, indicating that the additional topic modeling did not yield substantial enhancements in this context. The outcomes of the prediction accuracy, as detailed in Table 7, are further visualized in the form of a diagram, enhancing the comprehensibility of the analysis results. The graphical representation of the analysis outcomes is illustrated in Figure 7. Figure 7shows that Random Forest classifier achieved the highest accuracy (83%) and Gaussian Naive Bayes consistently demonstrated strong performance (80%) across feature representations, showcasing its robustness in handling the complexity of unsafe behavior prediction. On the other hand, Decision Trees and k-Nearest Neighbors displayed varied accuracy, suggesting a sensitivity to feature representations.

This case study highlights that the Decision Tree (DT) classifier exhibited the lowest accuracy compared to other classifiers among five models tested. The findings underscore the challenges faced by DT when confronted with increasing feature text lengths using BOW, TF-IDF, and GloVe. Algorithms such as BOW and TF-IDF generate sparse matrices with many zero-valued features, which can hinder DT's ability to effectively partition and classify data in high-dimensional, sparse spaces. Moreover, DT lacks the inherent capability to capture semantic relationships between words. Additionally, DT tends to become overly complex and deep when dealing with numerous features, leading to potential overfitting. These factors collectively explain DT's reduced performance in predicting the potential adverse effects of unsafe behaviors during hoisting operations in this study. It is noteworthy that the SBERT embedding showed an improved performance compared to the first case study, achieving 80% accuracy with the Random Forest classifier. This enhancement is attributed to the more detailed, context-rich textual descriptions utilized in this case.

In this case, zero-shot NLI pipeline achieved 35 % accuracy, 24 % precision, 35 % recall, and a 26 % F1-score. It perfectly retrieved all "Worker Harm" (WH) cases (recall = 1.00, F1 = 0.44), performed moderately on "Worker Injury" (WI) (precision = 0.44, recall = 0.40, F1 = 0.42), and failed entirely to identify any "Worker Death" (WD) instances (precision = recall = F1 = 0). These results highlight that generic, natural-language hypotheses ("This scenario would cause…") often miss the nuanced, domain-specific signals needed to distinguish high-severity outcomes, especially in a small dataset (n = 23).

6. Discussions

The literature models employed evaluation metrics, including accuracy, precision, recall, and F1-measure, to provide a comprehensive assessment of the classification model's performance. These widely recognized metrics were chosen to ensure a thorough and meaningful evaluation of the model's effectiveness in the context of the study. Table 8 demonstrates the classification outcomes derived from the conducted models. To further analyze the performance of our models, we conducted a comparative assessment with the results of existing model documented in the literature. This comparison enabled a thorough evaluation of how well our models performed relative to established benchmarks, offering valuable insights into their strengths and areas for potential improvement.

In the examination of predictive models within the chemical case study, where the dataset exhibited a balanced distribution, the Bag of Words (Ekramipooya et al., 2023) model emerged as the most effective, achieving a notable accuracy of 92%. Following closely, the GloVe model demonstrated a commendable performance with an accuracy of 89%. In contrast, for the hoisting operations dataset, the GloVe model outperformed other approaches, delivering the highest accuracy of 83%.

BOW model (Ekramipooya et al., 2023) focused on causes within a chemical context, achieves noteworthy precision, recall, and F1-score standing at 95%, 91%, and 91%, respectively. On the other hand, we designed the same model for the classification of consequences within hoisting operations, achieving a much lower accuracy of 80%. Precision, recall, and F1-score metrics for this model are 90%, 80%, and 81%, respectively. It must be recognized that the pivotal distinction lies in the length of the feature texts. The lengthier textual content within infrastructure projects poses a considerable challenge which contributed to the observed decrease in model performance. The limitation of this model stems from the inherent nature of BOW representation, which neglects the contextual meaning of words, resulting in a loss of semantic information. By treating each word in isolation and assigning equal importance to all terms, BOW generates sparse and high-dimensional

Table 8
Classification Results of developed models and Literature Model (BOW case (1)).

Index	Model	Accuracy	Precision	Recall (%)	F1-Score (%)
1	BOW+ML classifiers, (Ekramipooya et al., 2023)	0.92	0.95	0.91	0.91
2	TF-IDF+ML classifiers case (1)	0.86	0.9	0.86	0.86
3	TF-IDF+ LDA +ML classifiers case (1)	0.64	0.81	0.84	0.65
4	GloVe + ML classifiers case (1)	0.89	0.83	0.89	0.85
5	SBERT + ML classifiers case (1)	0.57	0.57	0.60	0.54
6	BOW+ML classifiers case (2)	0.8	0.9	0.8	0.81
7	TF-IDF+ML classifiers case (2)	0.8	0.9	0.8	0.81
8	TF-IDF+ LDA +ML classifiers case (2)	0.8	0.9	0.8	0.81
9	GloVe + ML classifiers case (2)	0.83	0.89	0.83	0.83
10	SBERT + ML classifiers case (2)	0.80	0.67	0.80	0.72

feature vectors that complicate computational efficiency and model interpretability. Additionally, the BOW algorithm does not consider word order, which is critical for capturing the structure and nuances of natural language.

Moreover, models 2 and 3, utilizing TF-IDF demonstrate slightly lower accuracy at 86% and 64%, respectively. These variations indicate a divergence in the models' ability to accurately classify instances, with model 2 displaying a more balanced performance compared to model 3. The examination of these models within hoisting operations yielded relatively consistent metrics (accuracy and recall: 0.8, precision: 0.9, and F1-score: 0.81). The inclusion of Latent Dirichlet Allocation (LDA) in conjunction with TF-IDF in both case studies did not contribute significantly to enhancing the model's performance. This was largely due to sparse data, short text lengths, and the specialized and repetitive nature of the vocabulary in these contexts, which limited the effectiveness of the approach. It should be noted that while the TF-IDF model highlights the importance of certain words over others, it does not address the challenges of high dimensionality and sparsity, nor does it utilize semantic similarities between words.

Model 4, employing GloVe embeddings and ML classifiers, achieves an accuracy of 89%, surpassing both models 2 and 3 in the chemical case study. Its precision and recall stand at 83% and 89%, respectively, resulting in an F1-score of 85%. This suggests that the GloVe embedding approach contributes positively to the model's overall performance, striking a favorable balance between precision and recall. GloVe model 9, in hoisting operations, stands out with an accuracy of 83%, showcasing a slightly higher performance compared to other models in same case, which all exhibit an accuracy of 80%. While GloVe maintains a precision of 89%, its recall and F1-score are both at 83%. GloVe method effectively addresses the limitation of capturing semantic relationships between words encountered in BOW and TF-IDF models by using a co-occurrence matrix. It is worth noting that the success of the GloVe models in both case studies is due to the carefully selected parameters. In particular, the relatively short sentences and closely related word pairs in the data points allowed GloVe's window size and vector dimensionality to effectively capture the relevant contextual relationships between data. This combination of factors enabled GloVe to outperform traditional models, providing more meaningful representations of the technical language used in HAZOP reports.

Table 8 shows that the GloVe algorithm's results in the second case study (Model 9) are lower than those in the first (Model 4) because the first case has evenly distributed, balanced data, making it easier for models to perform well. Additionally, the first case has fewer features (just causes), while the second case includes more complex features, including node names, guide words, parameters, deviations, causes, and countermeasures. This added complexity makes the model's task harder. We have already applied text preprocessing techniques such as removing stop words and punctuation as well as lemmatization and used GloVe with fixed 100-dimensional vectors to simplify the data. However, further complexity reduction could be achieved by using domain-specific pre-trained word embeddings to better capture specialized terms, applying dimensionality reduction techniques like PCA or t-SNE to compress high-dimensional embeddings while retaining key information, or incorporating Kolmogorov complexity and its features.

Figure 8 shows that BOW model (1) emerges as the top performer with short feature text length in terms of accuracy, precision, recall and F1-Score, while GloVe model follows closely, outperforming TF-IDF, TF-IDF+LDA, and SBERT models. SBERT exhibited the lowest performance in the first case study, yet showed improved accuracy in the second case. This improvement can be attributed to longer, more context-rich feature descriptions that enable SBERT to better capture semantic relationships at both word and sentence levels. Notably, GloVe achieved 83% accuracy, recall, and F1-score in the second case, outperforming other models, highlighting the robustness of simpler models in low-resource

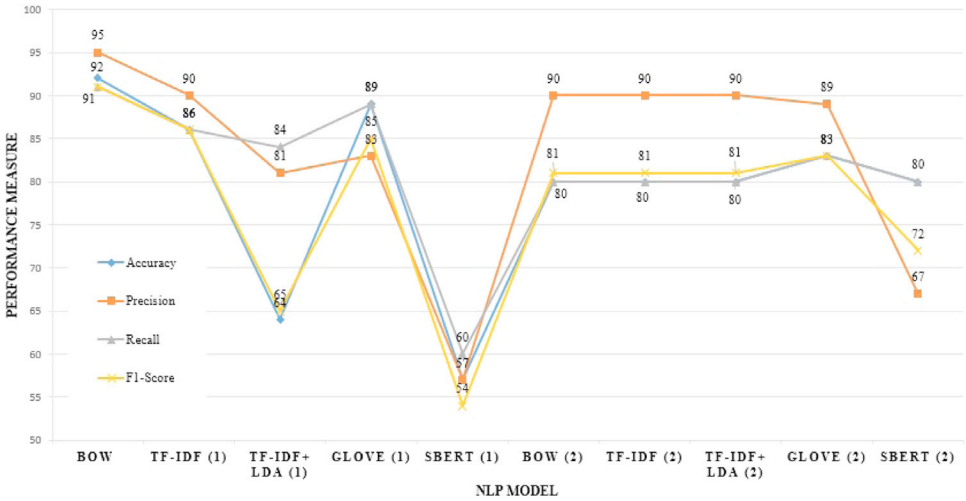


Figure 8. Classification Results of models compared to Literature.

settings. These findings underscore the importance of feature representation and selection in influencing the efficacy of text classification models. The efficacy of GloVe in capturing semantic relationships between words and its demonstrated performance indicates its applicability as a valuable tool for enhancing safety compared to other algorithms.

Across both studies—zero-shot NLI achieved moderate overall accuracy (52 % for deviations, 35 % for consequences) and good F1-score on common labels (e.g. “More Flow,” “No or Low Flow,” “Worker Harm”) but completely missed several under-represented, jargon-heavy classes (e.g. High Level, High Pressure, Low Level, Worker Death; F1-score = 0). These results stem from using a generic NLI model with simplistic hypothesis templates and testing on very small, datasets, which magnifies misclassification in rare categories. While zero-shot offers a quick, training-free bootstrap, it struggles with fine-grained, domain-specific distinctions. Future work will explore few-shot fine-tuning, enriched hypothesis phrasing, and hybrid pseudo-labeling to address these limitations.

The analysis of both case studies reveals that automating HAZOP reports for prefabricated building hoisting operations presents more challenges (complexity of longer text descriptions, diverse language, and limited data) compared to the chemical industry case study. Incorporating all HAZOP components—such as guide words, parameters, deviations, causes, consequences, and countermeasures—into the automation process can reduce model accuracy due to increased complexity. However, this approach better reflects the reality and interconnections between HAZOP elements in each instance. In the hoisting operations case study, the text features were longer and more detailed than in the chemical industry, encompassing various HAZOP components, which introduced additional complexity to the NLP tasks. Longer text descriptions make it more challenging for models to extract and accurately identify relevant information, leading to reduced performance. Furthermore, the language and context of HAZOP reports in this field are more diverse and influenced by a range of internal and external factors, making automation of hazard detection more difficult with NLP models typically designed for more structured, process-driven industries. Data limitations also played a significant role, as the hoisting operations dataset contained only 23 data points, limiting the ability to train robust models. In contrast, the chemical industry dataset was larger, offering more consistent data for model training. As a result, the models applied to the hoisting operations case study produced less favorablee outcomes compared to the chemical industry case, where shorter and more structured text allowed for better performance.

7. Conclusions and Recommendations

The traditional HAZOP process has long been recognized for its effectiveness in hazard identification and assessment; however, its time-consuming nature and resource-intensive report generation have prompted the necessity for automation. Over the past few years, the advent of advanced technologies, notably Natural Language Processing (NLP), have emerged as a transformative solution to streamline hazard assessment and reporting processes. In this research, we initiated the application of NLP strategies, starting with the chemical industry where strategies TF-IDF, TF-IDF+LDA, GloVe, and SBERT were employed to assess performance on datasets with shorter feature texts and compared with literature (BOW). A zero-shot NLI pipeline was further implemented to evaluate predicting labels without prior training. Subsequently, we extended our study to infrastructure projects, specifically in the domain of prefabricated buildings, where longer feature descriptions were prevalent. Through a comparative analysis with literature benchmarks, we found that the GloVe method achieved an accuracy exceeding 80% in this context. The research endeavors also encompassed the exploration of various ML classifiers in conjunction with NLP techniques. Among these, the Random Forest classifier notably stood out, exhibiting the highest accuracy when compared to other classifiers. The findings demonstrated that KNN performs poorly with short text features, as seen in the first case study, while DT struggles with long text features in NLP applications within HAZOP study, as shown in the second case study. The zero-shot model underscores low performance (52 % accuracy), lacking the precision needed for fine-grained, jargon-heavy labels. These outcomes underscore the significant potential of leveraging NLP methods, particularly GloVe, for the automation of HAZOP reports in operations of infrastructure projects. Such automation not only expedites the identification of hazards and operability issues but also offers a valuable resource for project managers in effectively managing and controlling safety concerns within infrastructure projects.

This work has some limitations that highlight directions for future research. The datasets, consisting of 42 datapoints in case study 1 and 23 in case study 2, are relatively small. Expanding these models to larger datasets from multiple HAZOP reports across industries would help validate the robustness and scalability of the methods. Additionally, domain-specific pretraining on industry data is also recommended to better capture technical terminology and improve predictive performance in HAZOP studies.

Another limitation is the low performance of the zero-shot classification model, which motivates a hybrid approach, integrating zero-shot and ML classifiers. Zero-shot pipeline is used to pre-label a subset of data, which is then manually refined and used to train ML models. This could improve performance while reducing the need for full manual labeling.

Additionally, Current prediction models rely solely on HAZOP reports, neglecting technical details from P&IDs, a critical input in HAZOP workshops for identifying deviations, causes, consequences, safeguards, and recommendations. Integrating historical P&IDs and HAZOP reports into prediction models is therefore essential. Leveraging AI tools such as object detection models to extract components and spatial information from P&ID nodes, combined with large multimodal models (LMM) to provide process and equipment descriptions, can enhance analysis. These tools, coupled with insights from associated HAZOP elements, can help generate an intelligent prepopulated HAZOP analysis table. This would guide participants during the workshop sessions by identifying whether the insights are sufficient or further deviations need consideration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Ehab Elhosary: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Osama Moselhi:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Data availability

The data utilized in this study, sourced from HAZOP reports, is contained within the paper text itself.

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